**Personal History:**

Yi Yin, Ph.D.
Assistant Professor
Department of Human Genetics
University of California, Los Angeles
Email: yeastyin@g.ucla.edu

**EDUCATION:**

| | | | |
|---|---|---|---|
| 2005 - 2009 | B.S. | Biotechnology | Beijing Normal University |
| 2011 - 2013 | Certificate | Computational Biology and Bioinformatics | Duke University |
| 2013 - 2015 | M.S. | Statistical Science | Duke University |
| 2009 - 2015 | Ph.D. | Genetics and Genomics | Duke University |

**PROFESSIONAL EXPERIENCE:**

| | |
|---|---|
| 2020 - | Assistant Professor, Department of Human Genetics, UCLA |
| 2016 - 2020 | Damon Runyon Postdoctoral Fellow, University of Washington (Advisor: Jay Shendure) |
| 2015 - 2016 | Damon Runyon Postdoctoral Fellow, Harvard University (Advisor: Sunney Xie) |
| 2009 - 2015 | Doctoral Student in Genetics and Genomics, Duke University (Advisor: Tom Petes) |
| 2013 - 2015 | Master Student in Statistical Science, Duke University (Advisor: Sayan Mukherjee) |
| 2009 | Undergraduate Student, National University of Singapore (Advisor: Stephen Cohen) |
| 2008 | Undergraduate Student, NIBS, China (Advisor: Xiaochen Wang) |
| 2006-2008 | Undergraduate Student, Beijing Normal University, China (Advisor: Yusheng Cong) |

**PROFESSIONAL ACTIVITIES:**

**Grant Review Services:**

| | |
|---|---|
| 2021 | Ad-hoc reviewer, Molecular Genetics B (MGB) Study Section, NIH |
| 2024 | Ad-hoc reviewer, Maximizing Investigators' Research Award A (MRAA) Study Section, NIH |
| 2025 | Reviewer, K99 Study Section, NIH |
| 2025 | Reviewer, Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) program Study Section, NIH |

**Editorial Services:**

| | |
|---|---|
| 2015 - | Ad hoc reviewer for Molecular Cell, Nature Methods, Nature Communications, Nucleic Acids Research, Cell Reports, Nature Genetics, PNAS, Mutation Research - Fundamental and Molecular Mechanisms of Mutagenesis, FEMS Yeast Research, Mutagenesis, Applied Microbiology and Biotechnology |

**Patent Applications:**

| | |
|---|---|
| 2025 | Yi Yin, Peter Chovanec. Sci-L3-Strand-seq. 2026-131: Sci-L3-Strand-Seq (Single-cell, 3-level combinatorial indexing for linear amplification and sequencing of replication template strands), UCLA Technology Development Group (TDG). |
| 2019 | Frank Steemers, Jay Shendure, Yi Yin. High-throughput single-cell sequencing with reduced amplification bias (US 62/821,864, PCT/US2019/032966, US 16/415,969; Publication of US20190382753A1) |
| 2018 | Frank Steemers, Jay Shendure, Yi Yin. High-throughput single-cell mapping of meiotic crossover and chromosome mis-segregation events (US 62/673,023, PCT/US2019/032966, US 16/415,969) |

**Professional Associations and Scholarly Societies:**

| | |
|---|---|
| 2026 | Session Chair, DNA Damage, Mutation and Cancer Gordon Research Conference |
| 2025 | Session Chair, The 2nd SCBA-Division of Genome Maintenance Scientific Symposium, Seattle, WA |

| | |
|---|---|
| 2025 | Session Chair and Panelist, Cell-Cell Symposium, Chan Zuckerberg Initiative, Los Angeles, CA |
| 2022 | Presentation of the Lifetime Achievement Award to Dr. Tom Petes, GSA Yeast Genetics Meeting, Los Angeles, CA |
| 2022 - 2026 | Program committee, GSA Yeast Genetics Meeting |
| 2022 | Session Chair, GSA Yeast Genetics Meeting 2022, UCLA |
| 2021 | Poster judge for the 5th SoCal Genome Stability Symposium, Scripps Institute, San Diego, CA (held virtually) |
| 2020 - 2021 | Member, American Association for Cancer Research |
| 2019 | Organizer of Career Development Workshops, FASEB, Genetic Recombination and Genome Rearrangements, Steamboat Springs, CO |
| 2013 - 2015 | Member, Genetics Society of America |

**Committee Service:**

| | |
|---|---|
| 2022 - 2025 | Legislative Assembly, UCLA |
| 2020 - | Graduate Student recruitment (reviewer for Bioinformatics IDP) |
| 2019 - | Graduate Student recruitment (reviewer for Genetics & Genomics IDP) |

**Teaching Service (Classroom):**

| | |
|---|---|
| 2025 | Instructor, Principles and Practices in Medical Genetics 1 (HG400A), UCLA |
| 2023 - current | Instructor, Bioinformatics Interdisciplinary Research Seminar (BIOINFO202 - Student Seminars), UCLA |
| 2021 - current | Instructor, Advanced Human Genetics B: Genetics and Genomics Aspects (HG236B), UCLA |
| 2022, 23 & 26 | Instructor, Ethics and Accountability in Biomedical Research (MIMG C234), UCLA |
| 2020 - 2025 | Instructor, Bioinformatics Interdisciplinary Research Seminar (BIOINFO202 - Faculty Seminars), UCLA |
| 2021 fall | Instructor, Seminar in Molecular Genetics (GGG295), UC Davis |
| 2017 spring | Instructor, Introduction to Computational Molecular Biology (GENOME541), University of Washington |
| 2013 fall | Teaching assistant, Genetic and Genomics Solutions to Biological Problems (UPGEN778), Duke University |
| 2012 fall | Teaching assistant, Genetic and Genomics Solutions to Biological Problems (UPGEN778), Duke University |

**Teaching Service (Mentoring):**

| | |
|---|---|
| 2020 - | Mentor for current Postdocs/Project Scientist - Peter Chovanec, Shiyang He, UCLA |
| 2021 - | Mentor for current PhD student - Trevor Ridgley (Bioinformatics IDP), UCLA |
| 2023 - | Mentor for rotation student - Vrishti Sinha, Richard Hu, Nabil Mohammed, Zhiyin Liu, Ziqi Rong, Yashas Appaji, Qingru Xu |
| 2021 - | Doctoral committee member - Aileen Nava (Valerie Arboleda lab), Seung Hyuk Tony Lee (Paivi Pajukanta lab), Vrishti Sinha (Caroline Kuo lab), Maris Kamalu (Steve Jacobsen lab), Natalia Garcia Dutton (Paul Spellman lab), UCLA |
| 2021 - | Mentor for undergraduate students - Jilai (Leo) Fan, Monica Lu, Michelle Ma, Stéphane Miaule, Jessica Javaherforoush, Chizaram (Zara) Okeiyi, Marat Zargaryan, UCLA; Junting Li, Shanghai Jiao Tong University, China (summer student at UCLA), Anusha Subramanian, UC Berkeley |
| 2020 - 2021 | Mentor for high school student - Renee Chang, Walnut High School (early admissions to MIT and Caltech) |
| 2019 | Postdoc mentor for rotation student - Shawn Fayer (Jay Shendure lab), University of Washington |
| 2015 - 2016 | Postdoc mentor for PhD student - Wenting Cai (Sunney Xie lab), Harvard University |
| 2013 | Student mentor for rotation student - Sarah Jaslow (Tom Petes lab), Duke University |

**Community Service:**

2023 Invited alumni guest and presenter, Molecular Genetics and Microbiology Departmental Retreat, Duke University, Wilmington, NC

2023 Invited career panelist, Mammalian DNA Repair Gordon Research Conference, Ventura, CA

2021 Invited panelist for the “STEM Fellowships 101 Workshop”, UCLA Postdoctoral Association

**Consulting Activities:**

2021 Collaborator: Wellcome Trust Senior Fellowship Award for Clinicians (PI: Nikhil Patkar, TATA Memorial Centre, India)

## HONORS AND SPECIAL AWARDS:

2026 Junior Faculty Award, W. M. Keck Foundation

2020 Damon Runyon Dale F. Frey Award for Breakthrough Scientists (Awarded to select Damon Runyon Fellows who have exceeded the highest expectations of their fellowship and demonstrate the greatest potential to become leaders in cancer research)

2019 Finalist (22/281 applicants), Burroughs-Wellcome Foundation (BWF) Career Award at the Scientific Interface

2016 - 2019 Damon Runyon Fellowship Award

2018 Outstanding Poster Presentation Award, Meiosis - Gordon Research Seminars (GRS)

2015 Certificate of Outstanding Contribution in Reviewing, Mutation Research-Fundamental and Molecular Mechanisms of Mutagenesis

2013 Chairman's Travel Awards, Duke University Medical Center

2008 First-class scholarship (top 3%), Beijing Normal University, China

2007 Second-class scholarship (top 6%), Beijing Normal University, China

2006 First-class scholarship (top 3%), Beijing Normal University, China

## RESEARCH GRANTS AND FELLOWSHIPS RECEIVED:

**Active:**

1002444 02/01/2026 - 01/31/2028 $500,000 (3 cal. mo.)

**W. M. Keck Foundation**: Complex recombination phenotyping at scale (PI: Yi Yin)

Purpose: mapping error-free and mutational homologous recombination (HR) in 50 cancer cell lines to understand how cancers with varied tissue origin differ in these two orthogonal signatures; examining both signatures in rare disease patient lines involving DNA repair gene knockouts.

R35 GM142511 08/01/2021 - 05/31/2026 $2,016,928 (6.1 cal. mo.)

06/01/2026 - 05/31/2027 No Cost Extension

**NIH (Impact Score: 18, ~2% based on program benchmarks)**: Understanding spontaneous mitotic crossover by single-cell multi-omics (PI: Yi Yin)

Purpose: applying state-of-the-art single-cell DNA sequencing technologies we developed to study mitotic crossover in yeast and mouse embryonic development.

**3R35GM142511-03S1 (Equipment Supplement)** 2023 $66,928

Supplement from NIH - National Institute of General Medical Sciences (PI: Yi Yin)

PC250408 09/01/2026 - 08/31/2030 $2,079,089 (3 cal. mo.)

**DOD CDMRP Idea Development Award (score 1.3, 5.41% applications funded):** Defining an Error-free DNA Repair Signature as a Predictive Marker for Lethal Prostate Cancer (PI: Yi Yin)

Purpose: defining an “HR” signature and refining the HR-deficient (“HRD”) mutation signature in localized and lethal prostate cancers, stratified by functional status of HR driver genes; and exploring a driver mutation-blind, genome rearrangement-informed biomarker for PARP inhibitor response (with co-I: Thomas Graeber, John K. Lee).

**Completed:**

DRG-2248-16 01/01/2016 - 12/31/2019 $224,000
**Damon Runyon** Fellowship: Global analysis of DNA break repair by single-cell sequencing (PI: Yi Yin)

DFS-43-20 04/01/2020 - 04/01/2022 $100,000
**Damon Runyon** - Dale F. Frey Award for Breakthrough Scientists (PI: Yi Yin)

## RESEARCH GRANTS PENDING AND OTHER RECOGNITION:

**Other Peer-Reviewed Grant Scores:**

R01 GM141239 04/01/2021 - 03/31/2026 $2,280,176
**NIH (Impact Score: 17, Percentile: 2%, MGB Study Section):** Development of a full-fledged mapping platform for mitotic crossover by single-cell multi-omics (PI: Yi Yin)
This is a focused technology development grant that has no overlap with other grants but was withdrawn due to institutional conflict with the NIGMS R35 funding mechanism. We proposed to develop single-cell sample multiplexing/hashing and *in vivo* sci-L3-Strand-seq/RNA co-assay.

**Pending:**

R35 GM142511 04/01/2027 - 03/31/2032 $2,327,890
**NIH**: Lesion-centric mechanisms of non-local DNA repair outcomes (PI: Yi Yin)
Purpose: dissecting the PARP1/2-HPF1 axis, testing two separable pro-amplification functions of HPF1 and the foldback model for palindromic amplification; asking whether post-replicative gaps from unexcised template lesions represent a mechanistically distinct non-local repair pathway independent of DSB-initiated repair; addressing why BRCA-null cancer lines but not normal fibroblasts exhibit high SCE, testing the roles of p53/DREAM-mediated repression, SLFN11 surveillance, and MMS22L-TONSL as a compensatory RAD51 loader, and uses reciprocal daughter cell pair signatures to dissect how converging-fork collapse at under-replicated regions may generate RAD51-independent SCE.

R01 CA319920 12/01/2026 - 11/30/2031 $3,039,061
**NIH**: Single-cell Genetic Analysis of Lesion-dependent Sister Chromatid Exchange Capacity with PARP Inhibition (PI: Yi Yin)
Purpose: determining how chronic endogenous *versus* acute pharmacologic PARP1/2 trapping induces sister chromatid exchange in various DNA repair gene knockouts or knockdowns, and identifying genetic regulators of this process; defining the mechanism of PARPi-induced, delayed sister chromatid exchange by testing whether it arises from *trans* cell cycle inheritance of unresolved ssDNA gaps or persistence of trapped PARP-chromatin complexes.

## LECTURES AND PRESENTATIONS:

**Invited Seminars:**

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|---|---|
| 2026 | Invited seminar, European Molecular Biology Laboratory (EMBL Heidelberg, Germany). |
| 2026 | DNA Dynamics Webinar Series (“Social DNAing”), Columbia University (New York, NY). |
| 2026 | Invited talk (30 invited speakers), DNA Damage, Mutation and Cancer Gordon Research Conference (Ventura, CA). |
| 2025 | Invited talk, The 2nd SCBA-Division of Genome Maintenance Scientific Symposium (Seattle, WA). |
| 2024 | Invited talk (30 invited speakers), DNA Damage, Mutation and Cancer Gordon Research Conference (Ventura, CA). |
| 2022 | Invited presentation of the Lifetime Achievement Award to Dr. Tom Petes, GSA Yeast Genetics Meeting (Los Angeles, CA). |

2022 Invited talk, GSA Yeast Genetics Meeting (Los Angeles, CA).
2021 Seminar in Molecular Genetics, UC Davis (Davis, CA).
2020 Bioinformatics-Human Genetics Seminar, UCLA (Los Angeles, CA).
2020 COVID-19 Basic, Translational and Clinical Research Seminar Series, UCLA (Los Angeles, CA).
2020 DNA Dynamics Webinar Series ("Social DNAing" - Rising Star), Columbia University (New York, NY).
2020 MBI Faculty Seminar, UCLA (Los Angeles, CA).
2019 National Institute of Environmental Health Sciences. Stadtman Tenure-Track Investigator Candidate Seminar (Durham, NC).
2019 National Institute of Aging. Stadtman Tenure-Track Investigator Candidate Seminar (Baltimore, MD).
2019 Molecular Biology Program, Sloan Kettering Institute, MSKCC (New York, NY).
2019 Department of Human Genetics Faculty Search Symposium. University of California, Los Angeles (Los Angeles, CA).
2017 Petes-Fest, Tom Petes 70$^{th}$ Birthday Symposium, Duke University (Durham, NC).

**Conference - Oral Presentations:**

2026 FASEB Genetic Recombination and Genome Rearrangements Meeting (Scottsdale, AZ).
2026 Cold Spring Harbor Asia Genome Integrity and Cancer Meeting (Suzhou, China).
2026 DNA Damage, Mutation and Cancer Gordon Research Conference (Ventura, CA).
2025 The 2nd SCBA-Division of Genome Maintenance Scientific Symposium (Seattle, WA).
2024 DNA Damage, Mutation and Cancer Gordon Research Conference (Ventura, CA).
2022 GSA Yeast Genetics Meeting (Los Angeles, CA).
2020 Genomic Stability and DNA repair, Virtual Keystone Symposia, (USA).
2019 Damon Runyon Fellows Retreat (Southbridge, MA).
2018 Illumina Sequencing Symposium (Seattle, WA).
2014 Duke University Department of Molecular Genetics and Microbiology Retreat (Wilmington, NC).
2011 Small Eukaryote Group Meeting, North Carolina Biotechnology Center (Durham, NC).
2011 Duke DNA Repair Group Meeting, Duke University (Durham, NC).

**Conference - Poster Presentations:**

2018 Molecular Mechanisms and Regulation of Meiosis Across Species. Gordon Research Conference & Seminar (New London, NH).
2017 DNA Replication and Recombination. Keystone Symposium (Santa Fe, NM).
2016 Damon Runyon Fellows Retreat (San Jose, CA).
2016 University of Washington Department of Genome Sciences Retreat (Leavenworth WA).
2013 Genetic Recombination and Genome Rearrangements, FASEB (Steamboat Springs, CO).
2012 Yeast Chromosome Structure, Replication & Segregation Conference, FASEB (Steamboat Springs, CO).

**PUBLICATION/BIBLIOGRAPHY:**

## RESEARCH PAPERS

RESEARCH PAPERS (PEER REVIEWED)

A. RESEARCH PAPERS - PEER REVIEWED (†: co-corresponding)

1. St Charles J, Hazkani-Covo E, **Yin Y**, Andersen SL, Dietrich FS, Greenwell PW, Malc E, Mieczkowski P, Petes TD. High-resolution genome-wide analysis of irradiated (UV and γ-rays) diploid yeast cells reveals a high frequency of genomic loss of heterozygosity (LOH) events. *Genetics.* 2012; 190(4): 1267-84. PMID: 22267500 PMCID: PMC3316642. (Role: generated and analyzed all the UV-induced LOH SNP-microarray data in this paper).

2. **Yin Y**, Petes TD. Genome-wide high-resolution mapping of UV-induced mitotic recombination events in S. cerevisiae. *PLoS Genet.* 2013; 9(10): e1003894. PMID: 24204306 PMCID: PMC3814309.

3. **Yin Y**, Petes TD. The role of Exo1p exonuclease in DNA end resection to generate gene conversion tracts in S. cerevisiae. *Genetics.* 2014; 197(4): 1097-109. PMID: 24835424 PMCID: PMC4125386.

4. **Yin Y**, Petes TD. Recombination between homologous chromosomes induced by unrepaired UV-generated DNA damage requires Mus81p and is suppressed by Mms2p. *PLoS Genet.* 2015; 11(3): e1005026. PMID: 25738287 PMCID: PMC4349867.

5. Deng SK, **Yin Y**, Petes TD, Symington LS. Mre11-Sae2 and RPA Collaborate to Prevent Palindromic Gene Amplification. *Mol. Cell.* 2015; 60(3): 500-8. PMID: 26545079 PMCID: PMC4636734. (Role: generated and analyzed the genome-wide mapping data in this paper).

6. **Yin Y**, Dominska M, Yim E, Petes T. High-resolution mapping of heteroduplex DNA formed during UV-induced and spontaneous mitotic recombination events in yeast. *eLife.* 2017; 6. pii: e28069. PMID: 28714850 PMCID: PMC5531827.

7. Guo X, Chavez A, Tung A, Chan Y, Kaas C, **Yin Y**, Cecchni R, Lopez-Garnier S, Kelsic E, Schubert M, DiCarlo J, Collins JJ, Church GM. High-throughput creation and functional profiling of DNA sequence variant libraries using CRISPR–Cas9 in yeast. *Nat Biotechnol.* 2018; 36(6): 540-546. PMID: 29786095 PMCID: PMC5990468. (Role: designed the pilot guide RNAs for this paper, and generated the yeast whole-genome sequencing data in the revision).

8. Chen W, McKenna A, Schreiber J, Haeussler M, **Yin Y**, Agarwal V, Noble WS, Shendure JA. Massively parallel profiling and predictive modeling of the outcomes of CRISPR/Cas9-mediated double-strand break repair. *Nucleic Acids Res.* 2019; pii: gkz487. 31165867 PMCID: PMC6735782. (Role: generated the monoclonal pCas9-BLAST 293T line and helped with data analysis and DNA repair mechanistic interpretation).

9. **Yin Y**†, Jiang Y, Lam KG, Berletch JB, Disteche CM, Noble WS, Steemers FJ, Camerini-Otero RD, Adey AC, Shendure JA†. High-throughput single cell sequencing with linear amplification. *Mol. Cell.* 2019; 76(4):676-690.e10. †Co-corresponding authors. PMID: 31495564 PMCID: PMC6874760.

10. Hilt EE, Boocock J, Trejo M, Le CQ, Guo L, Zhang Y, Sathe L, Arboleda VA, **Yin Y**, Bloom JS, Wang J, Elmore JG, Kruglyak L, Shrestha L, Mohamed Bakhash SA, Lin M, Xie H, Huang M, Roychoudhury P, Greninger A, Chandrasekaran S, Yang S, Garner OB. Retrospective

Detection of SARS-CoV-2 in Symptomatic Patients prior to Widespread Diagnostic Testing in Southern California. *Clinical Infectious Disease.* 2021; ciab360. PMID: 33939799 PMCID: PMC8135745. (Role: supervised the pilot SARS-CoV-2 viral genome sequencing experiments and continued to consult on the technical aspect of the sequencing).

11. Bloom JS, Sathe L, Munugala C, Jones EM, Gasperini M, Lubock NB, Yarza F, Thompson EE, Kovary KM, Park J, Marquette D, Kay S, Lucas M, Love T, Booeshaghi AS, Brandenberg OF, Guo L, Boocock J, Hochman M, Simpkins SW, Lin I, LaPierre N, Hong D, Zhang Y, Oland G, Choe BJ, Chandrasekaran S, Hilt EE, Butte M, Damoiseaux R, Kravit C, Cooper AR, **Yin Y**, Pachter L, Garner OB, Flint J, Eskin E, Luo C, Kosuri S, Kruglyak L, Arboleda VA. Massively scaled-up testing for SARS-CoV-2 RNA via next-generation sequencing of pooled and barcoded nasal and saliva samples. *Nat Biomed Eng.* 2021; Jul;5(7):657-665. PMID: 34211145 PMCID: PMC10810734. (Role: upon arrival at UCLA, established the strategic connection between the initial U.S. COVID-19 epicenter in Seattle to bridge UCLA Clinical Labs with the University of Washington's BBI - the first academic institute to test and sequence SARS-CoV-2. Initiated a communication network that synchronized technical operations between UCLA, UW, and Octant (Bay Area). Served as one of the five faculty to secure the initial $100k in seed funding from Jane Semel to launch the project. Imported technical expertise in the early planning of this project, including drafting IRB technical specifications and designing the protocols for the contrived sample validation of the technology).

12. Guo L, Boocock J, Hilt EE, Chandrasekaran S, Zhang Y, Munugala C, Sathe L, Alexander N, Arboleda VA, Flint J, Eskin E, Luo C, Yang S, Garner O, **Yin Y**[†], Bloom JS[†], Kruglyak L[†]. Genomic epidemiology of the Los Angeles COVID-19 outbreak and the early history of the B.1.43 strain in the USA. *BMC Genomics.* 2022; Apr 4;23(1):260. [†]Co-corresponding authors. PMID: 35379194 PMCID: PMC8978495. (Role: supervised all SARS-CoV-2 viral genome sequencing experiments from sample extraction to sequencing completion. Most of the samples were sequenced with V-seq, a rapid, cost-effective viral genome sequencing technology I developed for this project).

13. Chovanec PC, **Yin Y**. Generalization of the sci-L3 method to achieve high-throughput linear amplification for replication template strand sequencing, genome conformation capture, and the joint profiling of RNA and chromatin accessibility. *Nucleic Acids Res.* 2025 Feb 8;53(4):gkaf101. doi: 10.1093/nar/gkaf101. PMID: 39997216 PMCID: PMC11851118.

14. Chovanec PC, Ridgley TR, **Yin Y**. High-throughput mapping of spontaneous mitotic crossover and genome instability events with sci-L3-Strand-seq. *Nucleic Acids Res.* 2026 Feb 12;54(4):gkag119. doi: 10.1093/nar/gkag119. PMID: 41674384 PMCID: PMC12895072.

15. He S, Chovanec PC, **Yin Y**. Protocol for Mapping Sister Chromatid Exchanges and Structural Variations Using sci-L3-Strand-seq. *STAR Protocols*. 2026. Oct 7;7(4):104892. doi: 10.1016/j.xpro.2026.104892. PMID: 42848518 PMCID: PMC13676703.

16. Chovanec P, He S, **Yin Y**. Genome-wide mapping of Cas9-induced sister chromatid exchange across single and over 200 genomic targets. *eLife*. 2026 2026 Oct 12:15:RP112073. doi: 10.7554/eLife.112073. PMID: 42861852 PMCID: PMC13657629.

B. RESEARCH PAPERS - PEER REVIEWED (IN PRESS)

none

C. RESEARCH PAPERS - PEER REVIEWED (SUBMITTED)

1. Guo L, Boocock J, Tome JT, Chandrasekaran S, Hilt EE, Zhang Y, Sathe L, Li X, Luo C, Kosuri S, Shendure JA, Arboleda VA, Flint J, Eskin E, Garner OB, Yang S, Bloom JS†, Kruglyak L†, **Yin Y**†. Rapid cost-effective viral genome sequencing by V-seq. †Co-corresponding authors. *bioRxiv*, doi: doi.org/10.1101/2020.08.15.252510. (Minor Revision at *G3*).

2. Chovanec P, He S, **Yin Y**. Lesions initiating spontaneous mitotic crossover are minimally subject to non-homologous end joining. *bioRxiv* doi: doi.org/10.64898/2026.05.11.724412. (Minor Revision at *Nucleic Acids Res.*).

3. Collier A, Chovanec P, Yue S, Krall A, Heinkele F, Sha J, Ashley I, Sims R, Rubbi L, Feng B, Langerman J, Hwang Y, Lee M, Clark A, Pellegrini M, Christofk H, **Yin Y**, Bouhaddou M, Wohlschlegel J, Jonas S, Plath K. Pluripotent state-specific mechanisms orchestrate human blastoid development. (Role: supervised Chovanec P from my lab, 2nd author, to apply sci-L3-Strand-seq in reprogramming, and found that CNVs increase during naive cell reprogramming). (In Revision at *Nat. Cell Biol.*).

4. Ridgley T, Chovanec P, He S, **Yin Y**. Robust Chromosome Segmentation in Sparse Single-Cell Whole-Genome Sequencing Using a k-Segment Constrained Hidden Markov Model. *bioRxiv* doi: doi.org/10.64898/2026.10.19.760240. (Submitted to *Nucleic Acids Res.*).

D. RESEARCH PAPERS (NON-PEER REVIEWED)

none

E. RESEARCH PAPERS - NON-PEER REVIEWED (IN PRESS)

none

F. RESEARCH PAPERS - NON-PEER REVIEWED (SUBMITTED)

none

## CHAPTERS

1. Chovanec P, **Yin Y**. A mapping platform for mitotic crossover by single-cell multi-omics. *Methods Enzymol.* 2021;661:183-204. doi: 10.1016/bs.mie.2021.08.017. Epub 2021 Sep 11. PMID: 34776212 PMCID: PMC8944907.

CHAPTERS (IN PRESS)
none

## LETTERS TO THE EDITOR

none

## REVIEW

1. Klein HL, Bačinskaja G, Che J, Cheblal A, Elango R, Epshtein A, Fitzgerald DM, Gómez-González B, Khan SR, Kumar S, Leland BA, Marie L, Mei Q, Miné-Hattab J, Piotrowska A, Polleys EJ, Putnam CD, Radchenko EA, Saada AA, Sakofsky CJ, Shim EY, Stracy M, Xia J, Yan Z, **Yin Y**, Aguilera A, Argueso JL, Freudenreich CH, Gasser SM, Gordenin DA, Haber JE, Ira G, Jinks-Robertson S, King MC, Kolodner RD, Kuzminov A, Lambert SA, Lee SE, Miller KM, Mirkin SM, Petes TD, Rosenberg SM, Rothstein R, Symington LS, Zawadzki P, Kim N, Lisby M,

Malkova A. Guidelines for DNA recombination and repair studies: Cellular assays of DNA repair pathways. *Microb. Cell.* 2019 Jan 7;6(1):1-64. Review. PMID: 30652105; PMCID: PMC6334234.

**EDITORIALS**

none

**PAPERS IN PREPARATION (RESEARCH COMPLETED)**

1. Chovanec P*, He S*, **Yin Y**. Scalable and comprehensive mapping of crossover outcomes in rare disease patient-derived DNA repair gene variants reveals non-canonical sister chromatid exchange pathways.

2. Chovanec P*, He S*, **Yin Y**. Genetic regulation of error-free sister chromatid exchange and mutational structural variations induced by chronic endogenous *versus* acute pharmacologic PARP trapping.

3. **Yin Y**. Single-cell genome-wide analysis of UV-induced mitotic crossover in yeast *BLM* helicase knockouts revealed subclonal heterogeneity.

4. **Yin Y**. Single-cell genome-wide analysis of UV-induced mitotic crossover and whole-genome haploidization in yeast *POL32* knockouts.

**ABSTRACTS**

none